The 1+6+1 Partition of QED Selection Rules on S^3: Graph Topology, Angular Momentum, and the Dirac Spinor Phase Constraint

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Abstract

We classify the eight standard selection rules of perturbative QED according to which structural ingredient is required to recover them on the GeoVac graph. When QED is built on the scalar Fock graph with 1-cochain photons, only one of the eight rules survives—the Gaunt/Clebsch–Gordan sparsity built into the angular-momentum vertex. Building on the native Dirac graph with (n, κ, m_j) node labels recovers three additional spinor-intrinsic rules (Δm_j conservation, spatial parity, and Furry's theorem—the charge-conjugation consequence, recovered kinematically) for a total of 4/8. Promoting the photon from a scalar 1-cochain to a vector harmonic with explicit (q, m_q) angular labels and Wigner $3j$ vertex coupling recovers six angular-momentum rules at once (vertex parity, SO(4) channel count, the triangle inequality on n, Δm_j, spatial parity, Ward identity), at the cost of exactly one factor of $1/(4\pi)$ per loop, which we identify as the S^2 Weyl exchange constant of the Hopf base. This recovers 7/8 rules for scalar electrons. The eighth rule, Furry's theorem, is recovered for Dirac electrons by a kinematic identity of the single-particle vertex: the α-matrix is off-diagonal in the large/small-component decomposition, and the i phase on the small component reduces $V(a, a, q, m_q) = V_{LS} + V_{SL}$ to an imaginary part that m_l conservation forces to vanish identically. This is *not* the standard second-quantized C-symmetry mechanism; it is a single-particle phase constraint that produces Furry as a kinematic identity at the level of the vertex bilinear. The resulting partition is 1+6+1:

1. Graph-topological (Gaunt/CG sparsity), surviving on any GeoVac discretization regardless of photon promotion.

6. Angular-momentum, recovered by promoting the photon to a vector harmonic, costing exactly $1/(4\pi)$ per loop.

1. Dirac-kinematic (Furry), recovered by the single-particle Dirac spinor phase constraint, costing nothing (zero-cost identity at the vertex bilinear).

The partition is independent of the Paper 2 fine-structure-constant observation and stands on its own as an observation about the structural cost of recovering each QED selection rule on a discrete spectral construction. All claims are verified by 103 tests in `tests/test_vector_qed.py` (including aggregate tallies of all four census columns) plus the 21-test symbolic Furry derivation in `tests/test_paper33_furry_derivation.py`.

1 Introduction

Standard perturbative QED on flat space carries eight independent selection rules that constrain Feynman diagrams to nonzero amplitudes within their kinematic regimes. We list them in Table 1.

#	Rule	Mechanism in continuum QED
1	Gaunt / CG sparsity	$3j$ symbols and triangle inequality on $\lvert l_1 - l_2\rvert \leq l_\gamma \leq l_1 + l_2$
2	Vertex parity / GS structural zero	$l_a + l_b + l_\gamma$ odd at the $\bar{\psi}\gamma^\mu\psi A_\mu$ vertex
3	SO(4) channel count	W-coefficients of the Camporesi–Higuchi spinor decomposition vanish on disallow
4	Δm_j conservation	Magnetic quantum number addition rule $m_a + m_\gamma = m_c$
5	Spatial parity (E1)	Photon is a vector field, parity -1; matrix element vanishes between same-parity
6	Ward identity	$k_\mu \mathcal{M}^\mu = 0$, gauge invariance
7	Charge conjugation (Furry)	Closed fermion loops with odd number of vertices vanish under C
8	Triangle inequality on SO(4)	$\lvert n_1 - n_2\rvert \leq n_\gamma \leq n_1 + n_2 - 2$

Table 1: The eight standard selection rules of perturbative QED in continuum flat space. Each is enforced by a different structural ingredient.

When QED is constructed on the GeoVac graph (Paper 28 [10] §graph_native_qed, Sec. 2 below), only Rule 1 survives by default. Recovering each remaining rule requires a specific structural promotion of the construction. This paper documents which promotion recovers which rule, with what calibration exchange constant.

Why this matters. In the spectral-triple framework (Paper 32 [13]), QED on S^3 is the spectral action of the GeoVac Dirac operator D_{GV}. The eight selection rules are properties of that spectral action; each one corresponds to a specific structural feature of the underlying triple. Cataloguing which rules require which features clarifies which selection rules are *universal* (Paper 22 [7] sense, depending only on $\mathcal{A}$) and which are *Coulomb-specific* (Paper 31 [12] sense, depending on D). The 1+6+1 partition is the empirical answer.

Scope. This is an observation paper. The eight-rule table is standard; the GeoVac-specific contribution is the identification of which structural promotion recovers which rule, along with the per-loop $1/(4\pi)$ calibration cost and the zero-cost Dirac kinematic identity for Furry. No new physics, no fitting parameters; the result follows from explicit construction in `geovac/vector_qed.py` and verification in `tests/test_vector_qed.py` (103 tests, including aggregate tallies of all four census columns) and `tests/test_paper33_furry_derivation.py` (the symbolic Furry derivation, 21 tests).

2 The Graph-Native Baseline: 1/8

The graph-native QED construction of Paper 28 [10] §graph_native_qed builds all one-loop diagrams as finite matrix traces on the Fock graph. Photons are 1-cochains (functions on edges), photon propagator is the edge-Laplacian pseudo-inverse $G_\gamma = L_1^+$ where $L_1 = B^\top B$ is Paper 25's discrete Hodge-1 Laplacian, and the vertex is the bilinear $V(a,b,e) = \delta_{e,e_{ab}}$ enforcing that the two electron legs sit at the endpoints of the photon edge.

2.1 What survives

The only rule that survives this baseline is the Gaunt/CG sparsity of the angular-momentum vertex, which is built into the discrete edge set: the Fock graph has edges between (n, l, m_l) and (n', l', m_l') only if the standard ladder-operator selection rules are satisfied, which are exactly the Gaunt/CG triangle conditions for the $\bar{\psi}\psi$ matrix element of an angular-momentum-1 vector field. This is Rule 1 in Table 1.

2.2 What does not

Rules 2–8 fail on the graph-native baseline. In particular:

Proposition 1 (Pendant-edge theorem)**.** *Let $\Sigma_{\text{graph}}(\text{GS})$ denote the one-loop electron self-energy on the scalar Fock graph evaluated at the ground state $|1,0,0\rangle$. Then for every $n_{\max} \geq 2$,*

$$\Sigma_{\text{graph}}(\text{GS}) = \frac{2(n_{\max}-1)}{n_{\max}} \xrightarrow{n_{\max}\to\infty} 2.$$

In particular $\Sigma_{\text{graph}}(\text{GS}) \neq 0$ at every finite $n_{\max}$ and in the continuum limit, in stark contrast to the continuum QED self-energy structural zero $\Sigma_{\text{cont}}(n_{\text{ext}}=0)=0$ (Paper 28 [10] Theorem 4).

Mechanism. The ground state $|1,0,0\rangle$ is a pendant vertex (degree 1) of the Fock graph: its unique edge e_0 connects it to $|2,0,0\rangle$. The path-graph Laplacian inverse evaluated at this leaf edge gives $G_\gamma[e_0,e_0] = (n_{\max}-1)/n_{\max}$ exactly, and the vertex bilinear contributes a factor of 2; this is the only contribution to $\Sigma(\text{GS})$. Continuously verified in `tests/test_graph_qed_self_energy.py`: at $n_{\max}=2$ in exact rational arithmetic (the GS block is $[[1,1],[1,1]]$), and at $n_{\max}=3,4$ numerically to 10^{-10} (`TestPendantEdgeSweep`, added 2026-07-03 after the `graph_qed_photon` $n_{\max} \geq 4$ eigenvalue-sort crash fix; the exact-`sympy` pseudoinverse at $n_{\max} \geq 3$ exceeds any reasonable test budget, so those points are numeric). The closed form itself holds for all $n_{\max}$ by the path-graph Laplacian identity above. □

The continuum mechanism (vertex parity: $l_a + l_b + l_\gamma$ odd forces zero at the GS because both electron legs are $l=0$) does not act on the scalar 1-cochain photon, because the 1-cochain has no angular-momentum quantum number. This is the structural reason behind the pendant-edge theorem: the rule that should have produced the zero requires a structural feature (vector photon labels) that the construction does not have.

Conclusion. The graph-native baseline recovers 1/8 rules.

3 The Dirac Graph: 4/8

The native Dirac graph (CLAUDE.md §2, May 2026) replaces the scalar Fock node labels (n,l,m_l) with the relativistic Camporesi–Higuchi labels $(n_{\text{D}},\kappa,m_j)$, with two candidate adjacency rules: Rule A (κ-preserving) and Rule B (E1 dipole, $\Delta l = \pm 1$). Rule B recovers spinor-intrinsic selection rules that scalar Fock nodes do not carry.

3.1 What is recovered

Three additional rules are recovered:

- **(R4)** Δm_j conservation: built into the relativistic node label $m_j = m_l + m_s$, which is conserved at every electron–photon vertex with diagonal-in-m_j couplings.
- **(R5)** Spatial parity (E1): the parity of a Dirac spinor is $(-1)^{l(\kappa)}$ where $l(\kappa) = -1-\kappa$ for $\kappa<0$ and $l(\kappa)=\kappa$ for $\kappa>0$. Rule B adjacency $\Delta l = \pm 1$ enforces parity-changing transitions exactly as E1.
- **(R7)** Charge conjugation (Furry): the off-diagonal Rule B vertex makes the identity-vertex tadpole vanish on the Dirac graph, which is a graph-level analog of Furry's theorem (the single-vertex closed loop is zero by edge-graph antisymmetry).

3.2 What is not

Four rules remain unrecovered on the Dirac graph alone:

- (**R2**) Vertex parity / GS zero: requires the photon to carry an angular-momentum label l_γ, which the scalar 1-cochain does not.
- (**R3**) SO(4) channel count: requires the photon W-coefficients of the Camporesi–Higuchi decomposition.
- (**R6**) Ward identity: requires gauge invariance under the $U(1)$ phase of the photon, which the 1-cochain photon has only in temporal gauge and only approximately.
- (**R8**) Triangle inequality: the graph adjacency is nearest-neighbor in n ($\Delta n = \pm 1$ only), but the continuum triangle allows $|n_1 - n_2| \le n_\gamma \le n_1 + n_2 - 2$ with $n_\gamma \ge 2$, exceeding the graph's reach.

Verdict. Dirac graph alone recovers 4/8 rules. Three of the unrecovered rules (R2, R3, R6) are vector-photon-required; the fourth (R8) is a triangle-inequality range issue specific to the graph metric.

4 Vector-Photon Promotion: 7/8 (Scalar) and 8/8 (Dirac)

We now promote the photon from a scalar 1-cochain to a vector harmonic with explicit angular-momentum labels.

4.1 Construction

Definition 1 (Vector photon modes)**.** *For a maximum photon angular momentum $q_{\max} \ge 1$, the vector photon modes are the triples (q, m_q) with $1 \le q \le q_{\max}$, $-q \le m_q \le q$. The photon propagator is $G_\gamma(q) = 1/[q(q+2)]$, the eigenvalue of the vector Laplacian on S^3 at angular momentum q.*

Definition 2 (Wigner-$3j$ vertex)**.** *For electron states $a = (n_a, l_a, m_a)$, $b = (n_b, l_b, m_b)$ and photon mode (q, m_q), the vertex coupling is*

$$V(a, b, q, m_q) = \langle l_a, m_a | l_b, m_b; q, m_q \rangle_{3j} \cdot \theta(l_a + l_b + q \text{ odd}) \cdot R(n_a, n_b, q), \tag{1}$$

where the $3j$ Clebsch–Gordan symbol enforces angular-momentum addition, $\theta(\cdot)$ is the parity step function (1 if true, 0 otherwise), and $R(n_a, n_b, q) = 1$ for the present analysis (we work at the spinor-amplitude level; radial integrals are absorbed into the $3j$).

4.2 Selection rules recovered for scalar electrons

Theorem 1 (Scalar 7/8)**.** *With Definitions 1–2, the vector-photon QED on the Fock graph satisfies seven of the eight selection rules in Table 1, namely Rules 1, 2, 3, 4, 5, 6, 8. Rule 7 (Furry's theorem) fails: the vertex $V(a, a, q, m_q)$ is nonzero for $l_a \ge 1$ when q is odd, allowing nonzero closed-fermion-loop amplitudes with odd vertex count.*

Mechanism by rule. (R1) Built into the $3j$ symbol on $\langle l_a; l_b, q \rangle$.

- (R2) The parity step $\theta(l_a + l_b + q \text{ odd})$ is exactly the vertex parity rule; at the GS $l_a = l_b = 0$, this forces q odd, which combined with the magnetic-quantum-number constraint $|m_q| \le q$ and the Hermiticity of Σ produces $\Sigma(\text{GS}) = 0$ exactly.

(R3) The triangle inequality on the $3j$ symbol forces $|l_a - l_b| \leq q \leq l_a + l_b$; combined with the vector photon's full SO(4) harmonic content, this reproduces the continuum channel-count constraint. Census check (de-tautologized 2026-07-03): the per-(n_a, n_b) channel count realized in the vertex-tensor support matches the parity + triangle closed form for every pair (verified at $n_{\max} = 2, 3$).

(R4) The $3j$ symbol enforces $m_a + m_q = m_b$.

(R5) Built into the parity step.

(R6) At $n_{\max} = 2$ the Ward identity holds exactly (commutator ratio $\|[\Sigma, H_0]\|/\|\Sigma\| < 10^{-12}$); at $n_{\max} = 3$ it degrades (measured ratio ≈ 3.1) because cross-n couplings break the commutator metric on the truncated basis — the same degradation the Dirac census discloses (Theorem 2). The census is evaluated at $n_{\max} = 2$. (Corrected 2026-07-03: this item previously claimed a $\sim 10^{-3}$ residual at $n_{\max} = 3$, which the production diagnostic does not reproduce.)

(R7) FAILS: $V(a, a, q, m_q)$ for $a = (n, l \geq 1, m)$ and q odd contains terms proportional to $\langle l, m | l, m; q, 0\rangle_{3j} \neq 0$, allowing closed loops with odd vertex count.

(R8) Verified on the vertex-tensor support: every realized coupling satisfies $|n_a - n_b| \leq q \leq n_a + n_b - 2$ (census check `8_triangle_on_n`, added 2026-07-03). The upper bound follows from the parity step, the l-triangle, and $l \leq n - 1$; the lower bound is contingent and holds at the tested truncations $n_{\max} = 2, 3$.

$\square$

4.3 Furry's theorem from the Dirac spinor phase constraint

Theorem 2 (Dirac 8/8)**.** *Replacing the scalar electron states with Dirac spinors (large/small two-component decomposition with the standard i-phase on the small component, $\psi = (f\Omega_\kappa, ig\Omega_{-\kappa})$) and the γ^μ-vertex coupling with the $\boldsymbol{\alpha}$-matrix form, the vertex bilinear satisfies $V(a, a, q, m_q) = 0$ identically. In particular Furry's theorem (Rule 7) holds on the Dirac graph + vector-photon construction, recovering 8/8 selection rules (census evaluated at $n_{\max} = 2$; the Ward-identity ratio degrades at $n_{\max} = 3$ — see the open questions in §VIII).*

Proof. **Step 1: $m_q = 0$ is forced.** For the diagonal matrix element $a = b$ we have $m_{j,a} = m_{j,b}$, and the Wigner-Eckart m-rule of Definition 2 gives $m_q = m_{j,b} - m_{j,a} = 0$. Only the spatial z-component of the $\boldsymbol{\alpha}$-matrix contributes; we set $\alpha_z = \begin{pmatrix} 0 & \sigma_z \\ \sigma_z & 0 \end{pmatrix}$.

Step 2: bilinear evaluation. With the Dirac spinor $\psi = (f\Omega_\kappa, ig\Omega_{-\kappa})^\top$ and real radial functions f, g,

$$\begin{aligned} \alpha_z \psi &= (ig\sigma_z\Omega_{-\kappa}, f\sigma_z\Omega_\kappa)^\top, \\ \psi^\dagger \alpha_z \psi &= f^*\Omega_\kappa^\dagger(ig\sigma_z\Omega_{-\kappa}) + (-ig^*)\Omega_{-\kappa}^\dagger(f\sigma_z\Omega_\kappa) \\ &= ifg\big[\Omega_\kappa^\dagger\sigma_z\Omega_{-\kappa} - \Omega_{-\kappa}^\dagger\sigma_z\Omega_\kappa\big]. \end{aligned}$$

Since σ_z is Hermitian, the bracketed quantity equals $M - M^*$ where $M = \Omega_\kappa^\dagger\sigma_z\Omega_{-\kappa}$, giving $\psi^\dagger\alpha_z\psi = -2fg \cdot \Im M$.

Step 3: $\Im M$ vanishes pointwise. Expand the spin-spherical harmonics in the standard Clebsch–Gordan basis, $\Omega_{j,l,m_j} = \sum_{m_s} \langle l, m_l; \frac{1}{2}, m_s | j, m_j\rangle Y_l^{m_l}(\theta, \phi)\chi_{m_s}$ with $m_l = m_j - m_s$. The Pauli matrix $\sigma_z = \mathrm{diag}(+1, -1)$ is real-diagonal in the χ_{m_s} basis and conserves m_s. Therefore

$$M(\theta, \phi) = \sum_{m_s = \pm 1/2} (\operatorname{sign} m_s)\langle l_\kappa, m_l; \tfrac{1}{2}, m_s | j, m_j\rangle\langle l_{-\kappa}, m_l; \tfrac{1}{2}, m_s | j, m_j\rangle Y_{l_\kappa}^{m_l\,*} Y_{l_{-\kappa}}^{m_l},$$

where $m_l = m_j - m_s$ is fixed by σ_z conservation and the Clebsch–Gordan coefficients are real. The product $Y_{l_\kappa}^{m_l\,*}(\theta,\phi)\, Y_{l_{-\kappa}}^{m_l}(\theta,\phi)$ has matched magnetic indices and the $e^{\pm i m_l \phi}$ phases cancel exactly, leaving

$$Y_{l_\kappa}^{m_l\,*} Y_{l_{-\kappa}}^{m_l} = N_{l_\kappa, m_l}\, N_{l_{-\kappa}, m_l}\, P_{l_\kappa}^{|m_l|}(\cos\theta)\, P_{l_{-\kappa}}^{|m_l|}(\cos\theta),$$

which is a real function of θ alone. Hence $M(\theta,\phi)$ is purely real, $\Im M \equiv 0$, and $\psi^\dagger \alpha_z \psi = 0$ pointwise on S^2.

Step 4: integral vanishes. Since the integrand is identically zero before integration, $V(a,a,q,m_q) = \int_{S^2} \psi^\dagger \alpha_z \psi \cdot Y_q^0(\theta,\phi)\, d\Omega = 0$ for every $q \geq 1$.

The proof is closed. Six concrete cases at $\kappa \in \{-1,+1,-2\}$ and $m_j \in \{\pm 1/2, +3/2\}$ are verified symbolically in `tests/test_paper33_furry_derivation.py` (ported 2026-07-03 from the archived sprint driver): the bilinear $V_{LS} + V_{SL}$ simplifies to exactly zero pointwise on S^2 (Step 3), $\Im M \equiv 0$ with $M \not\equiv 0$ (the mechanism of Remark 1), and the projected integral $V(a,a,q,m_q) = 0$ sympy-exactly in every case. The derivation is independent of the production vertex code, whose diagonal short-circuit it backs. □

Remark 1 (The mechanism is m_l conservation, not Hermiticity)**.** *The cancellation in Step 3 comes from* two *structural inputs: (i) σ_z conserves m_l (because σ_z acts as a real-diagonal in the spin sector, leaving the spatial harmonic unchanged), and (ii) the matched magnetic index m_l on both sides of the bilinear forces the $e^{\pm i m_l \phi}$ phases to cancel, producing a purely real θ-dependent function. Hermiticity of α_z alone does not suffice: it gives $V_{LS} + V_{SL} = 2i\,\Im(V_{LS})$, but to conclude that this vanishes we need the* additional *input that $\Im(V_{LS}) = 0$, which is the m_l-conservation argument above.*

Remark 2 (This is not standard C-symmetry)**.** *Theorem 2 produces Furry's theorem from a* single-particle *kinematic identity at the level of the vertex bilinear, not from the second-quantized charge-conjugation C-symmetry of the field theory. The standard continuum derivation of Furry uses the antiunitary C operation $\psi \to C\bar{\psi}^T$ on the second-quantized field; the present construction never invokes second quantization—the closed fermion loop with odd vertex count vanishes because each individual vertex in the loop carries a factor of $V(a_i, a_i, q_i, m_{q_i}) = 0$ at the diagonal momentum configuration. This is a stronger statement: every individual diagonal vertex is zero, not just the sum.*

5 The 1+6+1 Partition Theorem

Combining Sec. 2–4 gives the full classification.

Theorem 3 (1+6+1 partition)**.** *The eight standard QED selection rules of Table 1 partition into three structural tiers based on which ingredient of the GeoVac construction is required to recover them.*

Tier I (graph topology, 1 rule). *Rule 1 (Gaunt/CG sparsity) is recovered by the graph-native baseline alone, on either the scalar Fock graph or the native Dirac graph, with or without vector-photon promotion. The rule is built into the discrete edge set as a triangle-inequality constraint on the $3j$ symbol.*

Tier II (angular momentum, 6 rules). *Rules 2, 3, 4, 5, 6, 8 (vertex parity,* SO(4) *channel count, Δm_j conservation, spatial parity, Ward identity, triangle inequality on n) are recovered* simultaneously *by promoting the photon from a scalar 1-cochain to a vector harmonic with* *explicit (q, m_q) labels and Wigner-3j vertex coupling. This promotion costs exactly one factor of $1/(4\pi)$ per loop, identified below (Sec. 6) as the S^2 Weyl exchange constant of the Hopf base.*

Tier III (Dirac kinematic, 1 rule). *Rule 7 (Furry's theorem) is recovered by the Dirac spinor phase constraint (Theorem 2), which forces the diagonal vertex bilinear $V(a, a, q, m_q)$ to vanish identically: Hermiticity and the off-diagonal structure of $\boldsymbol{\alpha}$ reduce it to an imaginary part, and m_l conservation kills that imaginary part (Remark 1 — the mechanism is m_l conservation, not Hermiticity alone). The cost of this recovery is zero: the constraint is a kinematic identity at the single-particle level, not a calibration.*

The total cost of recovering all eight rules from the graph-native baseline is:

$$\mathit{cost}(8/8) = \underbrace{0}_{\textit{Tier I (free)}} + \underbrace{n_{\text{loops}} \cdot \tfrac{1}{4\pi}}_{\textit{Tier II (per loop)}} + \underbrace{0}_{\textit{Tier III (kinematic)}} = \frac{n_{\text{loops}}}{4\pi}.$$

Proof. By Theorems 1 and 2, together with Proposition 1 and Sec. 3. Verified numerically in `tests/test_vector_qed.py`: each of the four census totals 1/8, 4/8, 7/8, 8/8 is asserted as an aggregate per-rule tally (`TestCensusAggregates`, added 2026-07-03), and the Tier-III Furry mechanism is derived symbolically in `tests/test_paper33_furry_derivation.py`. □

Rule	Tier	Scalar Fock	Dirac graph	Vector photon (scalar)	Vector photon (Dirac)
1. Gaunt/CG	I	PASS	PASS	PASS	PASS
2. Vertex parity (GS zero)	II	FAIL	FAIL	PASS	PASS
3. SO(4) count	II	FAIL	FAIL	PASS	PASS
4. Δm_j	II	FAIL	PASS	PASS	PASS
5. Parity (E1)	II	FAIL	PASS	PASS	PASS
6. Ward	II	FAIL	FAIL	PASS	PASS
7. Furry	III	FAIL	PASS*	FAIL	PASS
8. Triangle on n	II	FAIL	FAIL	PASS	PASS
Total		1/8	4/8	7/8	8/8

Table 2: Selection-rule census across four GeoVac construction tiers. * On the Dirac graph alone, Furry holds because the off-diagonal Rule B vertex makes the identity-vertex tadpole vanish; this is a graph-level analog of Furry, structurally distinct from the single-particle Dirac spinor phase constraint of Theorem 2. Both mechanisms produce the same outcome (closed odd-vertex loop = 0) but only the spinor phase constraint extends to the vector-photon construction.

6 The $1/(4\pi)$ Calibration Exchange Constant

The Tier-II promotion costs exactly $1/(4\pi)$ per loop. We identify this as the S^2 Weyl exchange constant of the Hopf base.

Proposition 2 ($1/(4\pi)$ identification)**.** *The vector-photon $l = 1$ self-energy on S^3 at $n_{\max} = 2$ equals exactly* $1/(4\pi)$.

Proof. Direct computation in `geovac/vector_qed.py`: the $l = 1$ self-energy is

$$\Sigma_{l=1}(n_{\max} = 2) = \sum_{q,m_q} \frac{|V(a,b,q,m_q)|^2}{q(q+2)} = \frac{1}{4\pi}$$

when summed over the only allowed q at $n_{\max} = 2$ ($q = 1$ with $m_q \in \{-1, 0, 1\}$). Verified in test `test_scalar_l1_self_energy_is_one_over_4pi` (`tests/test_vector_qed.py`). □

6.1 Why $1/(4\pi)$

The $S^2 = S^3/S^1$ Hopf base has total surface area 4π (in unit-radius normalization). Wigner-$3j$ matrix elements integrated over the full angular range produce a factor of $1/(4\pi)$ as the solid-angle normalization of the Weyl Hodge–star operator on the 1-form sector of S^2 [9, 1]. Phrased in the spectral-action vocabulary of [2] (the vocabulary, not a theorem of that reference), the per-loop factor $1/(4\pi)$ is read as the a_2 heat-kernel coefficient of the vector Laplacian on S^2, appearing as the Hopf-base contribution at each gauge-loop integration in the S^3 spectral expansion — GeoVac's own identification.

Observation 1 (Per-loop $1/(4\pi)$ as Paper 18 calibration)**.** *The factor $1/(4\pi)$ enters at each loop in the vector-photon construction. It is the S^2 Weyl exchange constant of the Hopf base, in the sense of Paper 18 [6]: a calibration exchange constant introduced by the projection of the spinor bundle onto the Hopf base. It is not derivable from the Coulomb potential alone and not derivable from the graph topology alone; it is the structural cost of recovering the angular-momentum selection rules of continuum QED on a discrete spectral construction.*

7 Connections to Other Papers

7.1 Paper 18 (exchange constant taxonomy)

The 1+6+1 partition of Theorem 3 maps cleanly onto Paper 18's three-tier exchange-constant taxonomy:

- Tier I (Gaunt/CG) is *intrinsic*: it lives in the discrete graph structure regardless of any continuous-limit choice.
- Tier II (angular momentum, 6 rules, costing $1/(4\pi)$ per loop) is *calibration*: the $1/(4\pi)$ is exactly Paper 18's characterization of a calibration exchange constant introduced by embedding the discrete graph in the continuum.
- Tier III (Furry, Dirac kinematic, zero cost) is also *intrinsic*, but to the spinor structure rather than to the graph: the kinematic identity holds at every truncation of the Dirac graph independent of the calibration.

In Paper 18 language, six of the eight QED selection rules are calibration content; one is intrinsic to the graph; one is intrinsic to the spinor. This is a sharper version of the universal/Coulomb-specific partition of Paper 31 [12].

7.2 Paper 22 (angular sparsity theorem)

Paper 22's angular-sparsity theorem states that ERI density depends only on $l_{\max}$, not on the potential $V(r)$. In the language of the present paper this is the statement that Rule 1 (Gaunt/CG sparsity, Tier I) is universal: it does not depend on the choice of Dirac operator or even on the Coulomb structure. Theorem 3 extends this: in addition to the angular-sparsity rule that Paper 22 identified, six more rules can be recovered by promotion to vector photons, but at the cost of a calibration; only one selection rule is truly potential-independent.

7.3 Paper 28 (QED on S^3)

Paper 28's continuum QED computations assume all eight selection rules hold; the present paper documents which structural ingredients of the GeoVac construction enforce each one. The pendant-edge theorem (Proposition 1) is Paper 28's $\Sigma_{\text{graph}}(\text{GS}) \neq 0$ result; the vector-photon recovery of Rule 2 is Paper 28's restoration of the continuum $\Sigma(n_{\text{ext}} = 0) = 0$ structural zero.

7.4 Paper 32 (spectral triple synthesis)

In Paper 32's spectral-triple language, the present partition is the statement that:

- Tier I is enforced by the algebra $\mathcal{A}_{\text{GV}}$ of the GeoVac triple.
- Tier II is enforced by the rank-1 vector-bundle promotion of the photon to a 1-form section of the Hopf base, which is the inner fluctuation of D_{GV} in the abelian extension of Paper 25. The $1/(4\pi)$ is the heat-kernel a_2 coefficient of the vector Laplacian on the Hopf base.
- Tier III is enforced by the real structure J_{GV} on $\mathcal{H}_{\text{GV}}$ (Paper 32 Proposition **??** analog), which intertwines large/small components with the standard C-charge-conjugation phase.

The present paper is therefore the spectral-action selection-rule audit of the GeoVac triple constructed in Paper 32.

8 Open Questions

Q1. Does the $1/(4\pi)$-per-loop calibration extend to two and three loops? At one loop the calibration is $1/(4\pi)$ per photon-vertex pair and the Tier-II cost is exactly $1/(4\pi)$. At two loops, naive counting gives $1/(4\pi)^2$ per closed photon sub-loop, but Paper 28's three-loop $O(N^3)$-factorization shows non-trivial cross-channel structure that may modify this. Whether the per-loop counting holds exactly or only at leading order is open.

Q2. Do all six Tier-II rules cost the *same* factor $1/(4\pi)$? Theorem 3 groups Rules 2–6, 8 together as Tier II because they are all recovered by the same structural promotion (vector photon). But it is possible that individual rules carry sub-leading corrections that distinguish them at higher loop order. A track-scale verification at two loops would either consolidate Tier II or split it.

Q3. Is the Dirac spinor phase constraint the only zero-cost selection rule? Theorem 2 is, to our knowledge, the only zero-cost (kinematic) selection rule identified in the present construction. Whether other QED selection rules—in higher-rank extensions, or in the rank-2 Breit sector flagged in Paper 25 §VII.B and Paper 32 Q5—also follow from analogous spinor phase constraints is an open structural question.

Q4. Where does the partition sit on the universal/Coulomb spectrum of Paper 31? Paper 31 partitions GeoVac results into universal (algebra-only) and Coulomb-specific (operator-dependent). The present 1+6+1 partition has Tier I universal, Tier II calibration (Coulomb-specific to S^3 via the Hopf base), and Tier III intrinsic to the Dirac sector (operator-dependent but spinor-universal). The structural correspondence between the two partitions is clean for Tiers I and II but slightly subtler for Tier III, which is universal across spinor systems but not across non-spinor ones (Paper 24's HO triple has no analog).

9 Conclusion

We have classified the eight standard QED selection rules into a 1+6+1 partition based on which structural ingredient of the GeoVac construction is required to recover them: one rule is graph-topological (Tier I, free), six are angular-momentum and recovered together by vector-photon promotion at cost exactly $1/(4\pi)$ per loop (Tier II, calibration), and one (Furry's theorem) is recovered by a kinematic identity of the Dirac single-particle vertex bilinear at zero cost (Tier III).

The partition is structural and verified by 103 tests in `tests/test_vector_qed.py` (each census total asserted as an aggregate per-rule tally) together with the symbolic Furry derivation in `tests/test_paper33_furry_derivation.py`. It does not depend on the Paper 2 observation about the fine-structure constant; it stands as an observation about the cost of recovering each QED selection rule on a discrete spectral construction.

The Tier-II identification of $1/(4\pi)$ as the S^2 Weyl exchange constant of the Hopf base places the entire angular-momentum sector of QED inside Paper 18's calibration tier. The Tier-III identification of Furry's theorem as a single-particle Dirac spinor phase constraint, distinct from second-quantized C-symmetry, is a new observation that, to our knowledge, has not been documented in this form in the QED literature.

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